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Full Stack Development

PROJECT REPORT
InHabit - Hostel Grievance Management System

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1. Motivation for Selection of the Topic

Hostel residents frequently depend on informal communication channels to report maintenance, cleanliness, safety, electrical, plumbing and other campus-related issues. Such complaints can be difficult to track, prioritise and close transparently. InHabit was selected to provide a centralised digital grievance-management platform in which students can raise and monitor complaints while authorised staff can manage, assign and resolve them. The project also extends the original Java/DBMS mini project into a modern full-stack web application for the Flexi Credit Course.

2. Objectives of the Project

- Provide a responsive web interface for students, wardens/administrators and service staff.
- Implement secure role-based login and controlled access to project functionality.
- Allow users to create, search, view, vote on and track hostel grievances through their lifecycle.
- Provide administrative workflows for status updates and assignment of complaints to technicians.
- Provide a Lost & Found module for reporting items and processing claims.
- Demonstrate FCC concepts including HTML5, CSS3, JavaScript, React, Node.js, Express REST APIs, MongoDB/Mongoose, Git and deployment-oriented configuration.

3. Customer Base / Target Users

The primary customer base is the student and residential community of a college hostel. Students are the main end users who submit and monitor grievances, vote on existing issues and report lost or found items.
Wardens/administrators act as supervisory users who review complaints, change their status and assign work. Technicians/service personnel form the operational user group responsible for handling assigned issues. The same architecture can be adapted for other campuses, hostels, residential societies or institutional facilities.

4. Summary of the Project

InHabit is a full-stack Hostel Grievance Management System developed for the Flexi Credit Course. The application uses a React-based frontend to provide reusable components, state-driven interfaces and responsive pages. A Node.js and Express backend exposes REST APIs for authentication, grievances, voting, status management, assignment and Lost & Found operations. MongoDB with Mongoose is used as the persistence layer. The application follows a

client–server architecture in which the browser sends HTTP requests to the Express API and the backend validates requests, performs business operations and stores/retrieves data from MongoDB.
The major workflow starts with role-based authentication. A student can create a grievance with relevant details, view existing complaints and vote on issues. Administrators/wardens can review complaints, update their status and assign them for resolution. Lost & Found provides a similar report-and-claim workflow. The interface is organised into reusable React components and uses JavaScript event handling and state management for interactive behaviour. API endpoints can be demonstrated and verified independently using Postman. The project is structured for Git-based version control and can be configured for deployment using the included environment/deployment files.

5. FCC Technologies and Concepts Demonstrated

FCC Concept	Implementation in InHabit
HTML5 / CSS3	Semantic web structure, responsive layout, forms, navigation and component styling.
JavaScript	Event handling, asynchronous API calls, validation and client-side state/interaction.
React	Reusable components, hooks/state, role-based views and dynamic grievance/Lost & Found screens.
Node.js + Express	REST API server, routing, middleware, request validation and JSON responses.
MongoDB + Mongoose	Document-based persistence, schemas/models, queries and application data storage.
Git / Testing / Deployment	Repository-based development, API verification with Postman and deployment configuration for the web stack.

6. Project Demonstration Focus
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Repository: https://github.com/iamthersforever/InHabit

- Login and role-based access.
- Create, search and view grievances; demonstrate voting and status changes.
- Demonstrate administrator/warden assignment workflow.
- Demonstrate Lost & Found reporting and claim workflow.
- Show React frontend, REST API calls and MongoDB-backed data flow.
- Use Postman to demonstrate selected REST endpoints and explain the Git/deployment structure.
- The project is intended to be demonstrated as a functional full-stack application. The implementation keeps the original problem domain and core grievance-management idea of the Java/DBMS mini project, while the FCC edition presents the system through a web-based React frontend and a Node.js/Express/MongoDB stack. This separation allows the FCC assessment to focus on the full-stack technologies and concepts required across Units 1 to 5.